

Syed Abbas Bukhari, PhD

Staff Scientist, Oncogenomics Section, National Cancer Institute, National Institutes of Health

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RESEARCH SUMMARY

Computational and systems biologist whose research connects regulatory programs to cellular and phenotypic variation. Doctoral and postdoctoral work in comparative and single-cell neurogenomics provides the foundation for current pediatric oncology research. Leads integrative analyses, develops reproducible software, and works with experimental investigators to formulate testable mechanisms. Research interests include tumor cell states, gene regulation, therapeutic dependencies, and rigorous evaluation of clinical relevance.

PROFESSIONAL EXPERIENCE

Staff Scientist, Oncogenomics Section, National Cancer Institute, NIH

2024-present | Bethesda, Maryland | Supervisor: Javed Khan, MD

- Lead computational studies of pediatric tumor heterogeneity by integrating single-cell and bulk transcriptomes with molecular subtype, copy-number inference, developmental-state estimates, and clinical outcomes.
- Direct the rhabdomyosarcoma analysis from atlas construction and expression-program discovery through regulatory interpretation, bulk-cohort projection, survival analysis, and manuscript preparation.
- Develop transcription-factor and gene-regulatory-network analyses to interpret recurrent cell states and evaluate potential effects of proliferation, genomic dosage, stress, and tissue composition.
- Contribute computational leadership to a pan-pediatric project combining tumor expression, regulatory networks, functional dependencies, and pharmacologic evidence to prioritize candidate therapeutic vulnerabilities.
- Build reproducible R/Python workflows for high-performance computing; document analytical decisions and mentor colleagues in study design, statistical evaluation, and biological interpretation.

Postdoctoral Research Fellow, Carl R. Woese Institute for Genomic Biology, University of Illinois Urbana-Champaign

February 2020-February 2023 | Urbana, Illinois

- Performed transcriptional regulatory-network analysis for a published neurogenomic meta-analysis (Scientific Reports, 2020); coauthored a perspective on behavior-related gene-regulatory networks (PNAS, 2020).
- Contributed methodology to the single-cell honeybee brain study published in Nature Ecology & Evolution (2023) during the IGB Postdoctoral Research Fellow appointment.

Bioinformatics Consultant, York University, Canada (remote, concurrent)

2021–2023 | Concurrent remote consultancy

- Provided bioinformatics analyses for honeybee toxicology research, contributing to recent publications on honeybee health and physiology, including the Proteomics study (2026).

Postdoctoral Research Associate, Department of Integrative Biology, University of Illinois Urbana-Champaign

August 2018-February 2020 | Urbana, Illinois

- Developed single-cell and bulk transcriptomic analyses of social-behavior systems and cross-species comparative frameworks.

Research Associate, Carl R. Woese Institute for Genomic Biology, University of Illinois Urbana-Champaign

2013-2017 | Urbana, Illinois

EDUCATION

- PhD, Informatics, University of Illinois Urbana-Champaign, August 2018. Dissertation: Neurogenomics of social interactions in male threespined sticklebacks. Advisor: Alison M. Bell, PhD.
- MS, Bioinformatics, University of Illinois Urbana-Champaign, August 2012.
- BS, Bioinformatics, COMSATS Institute of Information Technology, Islamabad, Pakistan, April 2009.

CURRENT AND RECENT RESEARCH

Cell-state programs and clinical heterogeneity in rhabdomyosarcoma (2024-present)

Role: Computational lead. Status: analysis and lead-author manuscript in preparation.

- Integrated 321,704 cells from RMS, developing muscle, and adult muscle. After dissociation-stress filtering, the RMS discovery set contains 104,917 tumor-derived cells from 21 patients, including malignant and microenvironmental cells.
- Combined copy-number-informed cell classification and developmental-potency estimates with multi-rank per-sample NMF; cross-sample consensus identified 32 metaprograms for functional and regulatory interpretation.
- Projected fusion-matched programs into clinically annotated bulk cohorts using GSVA and evaluated outcomes with univariate Cox models, Kaplan–Meier analysis, proportional-hazards diagnostics, and multiple-testing correction. Developed SigClust for statistically supported annotation of existing single-cell clusters.
- Current outcome findings are exploratory and do not survive Benjamini–Hochberg correction. Interpretation accounts for limited events, follow-up limitations, and the distinction between expression enrichment and cell abundance.

Pan-pediatric GRN and druggable-dependency discovery (2024-present)

Role: Computational lead contributor. Status: manuscript in preparation.

- Constructed consensus gene-regulatory networks using ARACNe-AP and GENIE3 across pediatric solid tumors.
- Combined TF/regulon evidence and tumor-specific expression with DepMap dependencies and PRISM, ChEMBL, DrugBank, DrugCentral, and MIPE resources to rank candidate regulators.
- Organized evidence supporting candidate vulnerabilities, including potential confounding by proliferation, tissue composition, and generic essentiality; experimental validation remains a subsequent step.

Cross-species neurogenomics and behavior-related GRNs (2013-2023)

- Integrated time-resolved RNA-seq, H3K27ac ChIP-seq, and regulatory-network evidence in first-author work (PLOS Genetics, 2017).
- Tested competing transcriptional models and harmonized cross-species comparisons with sensitivity and permutation analyses (Nature Communications, 2019).
- Contributed methodology to the single-cell honeybee brain study (Nature Ecology & Evolution, 2023) and data analysis to comparative neurogenomics (PNAS, 2014).
- Compared gene-, pathway-, and network-level responses across species and experimental designs to distinguish shared biological organization from context-specific variation.

TECHNICAL EXPERTISE

- Computing and reproducibility: R, Python, Bash, Bioconductor, SLURM/Biowulf, Docker/Singularity, conda/mamba, Git, workflow design, provenance-aware reporting.
- Single-cell, bulk, and integrative genomics: Seurat, Scanpy, Harmony, CopyKAT, CytoTRACE2, SingleR, reference mapping, program scoring, RNA-seq/H3K27ac ChIP-seq integration.
- Networks and programs: ARACNe-AP, GENIE3, SCENIC/pySCENIC, DoRothEA, NMF, WGCNA, regulon activity, latent-factor models.
- Statistics and clinical modeling: GSVA/UCell, SigClust signature annotation, Cox regression, Kaplan–Meier analysis, proportional-hazards diagnostics, multiple-testing correction, resampling, and sensitivity analysis.

SOFTWARE

- SigClust: Open-source R workflow for signature-based annotation of pre-existing single-cell clusters using continuous per-cell scores, one-sided Fisher enrichment, odds ratios, and false-discovery-rate control. Produces cluster annotations, enrichment heatmaps, functional UMAPs, and machine-readable reports. <https://github.com/bukhariabbas/SigClust>
- POLYAR: Prediction of human transcription termination poly(A) sites; associated peer-reviewed publication in BMC Genomics.

PEER-REVIEWED PUBLICATIONS

1. Zhong H, Shi Y, Kozlova A, Moravcova R, Rogalski JC, Jamieson A, Lansing L, Moon K-M, Yuan X, Gregoris AS, Higo H, Common J, Conflitti IM, Pepinelli M, Tran L, Cunningham M, Jabbari H, Bukhari SA, et al. Omics insights into the effects of highbush blueberry and cranberry crop agroecosystems on honey bee health and physiology. *Proteomics*. 2026;26(8):41-57. doi:10.1002/pmic.70033.
2. Traniello IM, Bukhari SA, Dibaeinia P, Serrano G, Avalos A, Ahmed AC, Sankey AL, et al. Single-cell dissection of aggression in honeybee colonies. *Nature Ecology & Evolution*. 2023;7(8):1232-1244. doi:10.1038/s41559-023-02090-0.
3. Hellmann JK, Bukhari SA, Deno J, Bell AM. Sex-specific plasticity across generations I: maternal and paternal effects on sons and daughters. *Journal of Animal Ecology*. 2020;89(12):2788-2799.
4. Sinha S, Jones BM, Traniello IM, Bukhari SA, Halfon MS, Hofmann HA, Huang S, et al. Behavior-related gene regulatory networks: a new level of organization in the brain. *Proceedings of the National Academy of Sciences USA*. 2020;117(38):23270-23279. doi:10.1073/pnas.1921625117. (Perspective.)
5. Traniello IM, Bukhari SA, Kevill J, Ahmed AC, Hamilton AR, Naeger NL, Schroeder DC, Robinson GE. Meta-analysis of honey bee neurogenomic response links Deformed wing virus type A to precocious behavioral maturation. *Scientific Reports*. 2020;10:3101. doi:10.1038/s41598-020-59808-4.
6. Bukhari SA, Saul MC, James N, Bensky MK, Stein LR, Trapp R, Bell AM. Neurogenomic insights into paternal care and its relation to territorial aggression. *Nature Communications*. 2019;10:4437. doi:10.1038/s41467-019-12212-7.
7. Saul MC, Blatti C, Yang W, Bukhari SA, Shpigler HY, Troy JM, Seward CH, et al. Cross-species systems analysis of evolutionary toolkits of neurogenomic response to social challenge. *Genes, Brain and Behavior*. 2019;18(1):e12502.
8. Stein LR, Bukhari SA, Bell AM. Personal and transgenerational cues are nonadditive at the phenotypic and molecular level. *Nature Ecology & Evolution*. 2018;2:1306-1311. doi:10.1038/s41559-018-0605-4.
9. Bukhari SA, Saul MC, Seward CH, Zhang H, Bensky MK, James N, Zhao SD, Chandrasekaran S, Stubbs L, Bell AM. Temporal dynamics of neurogenomic plasticity in response to social interactions in male threespined sticklebacks. *PLOS Genetics*. 2017;13(7):e1006840. doi:10.1371/journal.pgen.1006840.
10. Bell AM, Bukhari SA, Sanogo YO. Natural variation in brain gene-expression profiles of aggressive and nonaggressive individual sticklebacks. *Behaviour*. 2016;153(13-14):1723-1743. doi:10.1163/1568539X-00003393.
11. Paitz RT, Bukhari SA, Bell AM. Stickleback embryos use ATP-binding cassette transporters as a buffer against exposure to maternally derived cortisol. *Proceedings of the Royal Society B*. 2016;283(1826):20152838.
12. Rittschof CC, Bukhari SA, Sloofman LG, Troy JM, Caetano-Anolles D, Ahmed AC, Kent M, Lu X, Sanogo YO, Weisner PA, Zhang H, Bell AM, Ma J, Sinha S, Robinson GE, Stubbs L. Neuromolecular responses to social challenge: common mechanisms across mouse, stickleback fish, and honey bee. *Proceedings of the National Academy of Sciences USA*. 2014;111(50):17929-17934. doi:10.1073/pnas.1420369111.
13. Bukhari SA, Caetano-Anolles G. Origin and evolution of protein fold designs inferred from phylogenomic analysis of CATH domain structures in proteomes. *PLOS Computational Biology*. 2013;9(3):e1003009. doi:10.1371/journal.pcbi.1003009.
14. Akhtar MN, Bukhari SA, Fazal Z, Qamar R, Shahmuradov IA. POLYAR, a new computer program for prediction of poly(A) sites in human sequences. *BMC Genomics*. 2010;11:646. (Equal contribution.)

MANUSCRIPTS IN PREPARATION

- Bukhari SA, et al. Cell-state programs and clinical heterogeneity in rhabdomyosarcoma. Lead-author manuscript in preparation, 2026.
- Bukhari SA, et al. Pan-pediatric regulatory networks and druggable dependencies across solid tumors. Manuscript in preparation, 2026.

CONFERENCE PROCEEDING

- Bukhari SA, Caetano-Anolles G. Evolution of protein architectures inferred from phylogenomic analysis of CATH. IEEE International Conference on Bioinformatics and Biomedicine Workshops, 2011; Atlanta, Georgia.

MENTORING, TEACHING, AND OUTREACH

- Mentor colleagues and junior researchers in single-cell and bulk transcriptomics, regulatory networks, survival analysis, and reproducible high-performance computing.
- Use written analysis plans, worked examples, and code review to develop independent computational reasoning and interpretation.
- Prior instructional roles: Teaching Assistant, Applied Statistical Methods I (more than two years); Teaching Assistant, Introduction to Applied Statistics (one semester); Grader, Molecular Genetics (one semester).
- Prior invited and guest lectures and hands-on computational-genomics instruction during UIUC training.

HONORS AND AWARDS

- HTAN Data Jamboree travel award (offered; travel supported through NIH government funds), 2025.
- Dissertation Completion Fellowship, University of Illinois Urbana-Champaign, 2017-2018.
- Travel Award, Big Data Neuroscience Workshop, Indiana University, 2017.
- Travel Award, 8th International Conference of Stickleback Behavior and Evolution, 2015.
- Travel Award, Annual Computational Genomics Workshop, Broad Institute/MIT, 2015.
- CIIT Fellowship, MS Bioinformatics, 2010-2012.
- Silver Medals (Institute and Campus), COMSATS Institute of Information Technology, 2009.

INVITED TALKS

- Neurogenomics of social behavior in male threespined sticklebacks. Ecology, Evolution and Behavior Seminar, University of Missouri, September 2019.
- Neurogenomic insights into paternal care. IGB Fellows Symposium, May 2019.
- Neurogenomics of social interactions in male threespined sticklebacks. University of Texas at Austin, January 2018.

SELECTED OTHER PRESENTATIONS

- Temporal dynamics of neurogenomic response to a social challenge in male threespined stickleback. 8th International Conference of Stickleback Behavior and Evolution, July 2015.
- Shared socially responsive gene families between sticklebacks and fruit fly. IGB Gene Networks and Developmental Plasticity Theme Data Blitz, June 2014.
- Temporal dynamics of neurogenomic response to a territorial challenge in male threespined sticklebacks. RCN Meeting, 2015.
- Socially responsive gene families conserved across social interactions in fruit fly and stickleback fish. Gordon Research Conference, 2014.

PROFESSIONAL SERVICE

- Peer reviewer for computational and systems-biology manuscripts.
- Ongoing mentorship and analytical consultation within the NCI Oncogenomics Section.